PASSPROOF – SMART OD MANAGEMENT SYSTEM

CS23531 – WEB PROGRAMMING MINI PROJECT REPORT

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The report describes the project concept, system design, selected technologies, and implementation approach.

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ABSTRACT

The management of On-Duty (OD) requests in many engineering colleges still relies on paper-based methods, resulting in delays, excessive paperwork, inefficient approval workflows, and difficulty in verifying the authenticity of student OD claims. These challenges cause communication gaps among students, faculty, and departmental authorities and increase the possibility of fraudulent OD usage.

PassProof – Smart OD Management System is a secure web-based application developed as a Web Programming mini project that automates the complete OD management process. The platform supports six user roles — Administrator, Student, Event Head, Department Head, Head of the Department (HOD), and Faculty — each with a dedicated dashboard and role-specific permissions. Students submit Individual OD requests and Event Heads submit Event OD requests, which pass through a two-level approval workflow involving the Department Head and the HOD.

Each approved request automatically generates a digitally verified OD Pass with a unique QR Code, which faculty can validate instantly by scanning the code or searching the student's register number. The system also provides bulk user upload through Excel, real-time status tracking, notifications, conflict and duplicate-request detection, audit logging, PDF pass generation, and analytics dashboards.

The frontend is built using HTML5, CSS3, and JavaScript, the backend using Python (Flask), and data is stored in MongoDB. JSON Web Token (JWT) authentication with Role-Based Access Control (RBAC) and bcrypt password hashing ensure secure access. This report documents the project concept, requirements, system architecture, module descriptions, database design, implementation screenshots, security features, and testing outcomes. Overall, PassProof offers a scalable, secure, and user-friendly solution that modernizes institutional OD management.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO.
1	Introduction	1
2	Purpose and Scope	2
3	Software Requirement Specification	3
4	System Design	4
5	Module Description	5
6	Database Design	6
7	Implementation	7
8	Security Features	11
9	Testing	11
10	Conclusion and Future Enhancements	12
11	Sustainable Development Goals (SDG) Mapping	13
12	References	13
	Appendix	13

1. INTRODUCTION

1.1 Project Overview

Digital transformation has improved academic and administrative activities in educational institutions, yet On-Duty (OD) requests in many engineering colleges still follow paper-based procedures. Students taking part in symposiums, workshops, internships, industrial visits, placement drives, and sports competitions must obtain handwritten approvals and manual verification, which is slow and creates communication gaps.

PassProof – Smart OD Management System is a web-based application that automates the complete OD approval process. Students submit OD requests digitally, and the Event Head, Department Head, and Head of the Department (HOD) review them through a secure workflow. Once approved, the system generates a digitally verified OD Pass with a unique QR Code that faculty can verify instantly.

The application uses Python Flask (backend), MongoDB (database), and HTML5, CSS3, and JavaScript (frontend). Authentication is handled with JSON Web Tokens (JWT) for role-based access control. Conflict detection, audit logging, analytics dashboards, PDF generation, and notifications further reduce administrative workload.

1.2 Problem Statement

Institutions depend on manual methods where students submit paper applications and collect approvals from several authorities, consuming time and causing delays. Faculty find it difficult to verify whether an absence is backed by a valid OD, and manual records make retrieval, monitoring, and reporting hard. Paper documents can be lost, damaged, duplicated, or altered. Without a central platform, students are unaware of their application status and administrators cannot easily monitor pending requests, which shows the need for an automated, secure, and transparent OD management solution.

1.3 Existing System

In the existing system, students obtain OD forms, attach supporting documents, and visit several faculty members for handwritten approvals, which are later checked manually. Its limitations are:

- Paper-based documentation increases workload and approvals are slow.
- No centralized repository; records are hard to retrieve or report on.
- Faculty cannot verify OD requests instantly.
- Documents may be lost or duplicated, and fraudulent claims are possible.
- No transparency in approval status and limited student–authority communication.

1.4 Proposed System

PassProof digitizes the complete OD workflow. Students submit Individual OD requests and Event Heads submit Event OD requests from their dashboards. Each application goes through two-level approval by the Department Head and the HOD. After approval, a unique digital OD Pass with a QR Code is generated, which faculty can verify by scanning the code or searching the register number. Audit logs, analytics, notification history, and approval timelines give complete transparency.

1.5 Objectives

- Automate the complete On-Duty approval process and reduce paperwork.
- Provide secure role-based authentication and support both Event OD and Individual OD.
- Generate QR Code-enabled OD passes that faculty can verify instantly.
- Maintain centralized records with real-time tracking and notifications.
- Generate analytical reports to improve transparency, accountability, and efficiency.

2. PURPOSE AND SCOPE

2.1 Purpose of the Project

The purpose of PassProof is to provide a secure, efficient, and centralized digital platform for managing OD requests, replacing paper-based procedures with an automated workflow that improves transparency and communication among students, faculty, and authorities. Every request passes through the Department Head and the HOD before a digitally verified OD Pass is generated. Unique QR Code passes help eliminate fraudulent claims, while secure authentication, role-based access control, audit logging, and conflict detection strengthen the integrity of the system.

2.2 Scope of the Project

The scope covers the complete lifecycle of OD management, from application submission to digital verification by faculty. Students apply for Individual OD with supporting documents; Event Heads create events and submit Event OD for participants; Department Heads give first-level approval and the HOD gives final approval. Faculty verify passes by QR scan or register number. The system also provides user management, bulk upload, analytics, report generation, and duplicate-request detection, with a responsive interface for desktops, tablets, and mobiles.

The current implementation targets engineering colleges, but the architecture is scalable to universities and other organizations. Future enhancements include cloud deployment, mobile apps, biometric integration, digital signatures, AI-based analytics, and ERP integration.

3. SOFTWARE REQUIREMENT SPECIFICATION

The SRS defines the functional and non-functional requirements, hardware and software needs, and user roles of PassProof, ensuring the system meets the needs of students, faculty, administrators, and departmental authorities with security, scalability, and reliability.

3.1 Functional Requirements and User Roles

The system supports six user roles. Each authenticated user can access only the features permitted for their role.

User Role	Responsibilities and Key Functions
Administrator	Create and manage user accounts, bulk upload students and staff using Excel, activate/deactivate accounts, reset passwords, monitor system usage.
Student	Submit Individual OD requests, upload supporting documents, track approval status, download QR-based OD Pass, view OD history.
Event Head	Create events, add participating students, submit Event OD requests, monitor approval status, update event details before submission.
Department Head	First-level approval: review pending requests, verify documents, approve/reject, forward to HOD, view approval history.
Head of Department	Final approval: review requests, upload official approval letter, approve/reject, generate QR-enabled OD Pass, view analytics.
Faculty	Search student by Register Number, scan QR Code, verify digital OD Pass, view approval information and daily OD status.

3.2 Non-Functional Requirements

Category	Requirements
Performance	Fast login and approval response, quick QR verification, efficient database queries.
Security	JWT-based authentication, role-based authorization, bcrypt password storage, secure document upload.
Reliability	Availability during college working hours, reliable OD record storage, automatic recovery from failures.
Scalability	Supports thousands of users, expandable database, easy addition of future modules.
Usability	Simple, responsive, mobile-friendly interface with easy navigation.
Maintainability	Modular backend, well-structured source code, reusable components, easy database maintenance.

3.3 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above
RAM	4 GB or higher
Hard Disk	500 GB or higher
Operating System	Windows 10/11, Linux, macOS
Internet	Broadband Connection
Display	1366 × 768 Resolution

3.4 Software Requirements

Category	Software / Technology
Programming Language	Python 3.x
Backend Framework	Flask
Frontend	HTML5, CSS3, JavaScript
Database / Driver	MongoDB / PyMongo
Authentication	Flask-JWT-Extended
Password Encryption	bcrypt
QR Code / PDF Generation	qrcode / ReportLab
Excel Processing	openpyxl, pandas
IDE / Browser	Visual Studio Code / Google Chrome

4. SYSTEM DESIGN

This section shows the overall structure of PassProof, how users interact with it, and how information flows between its components.

4.1 System Architecture

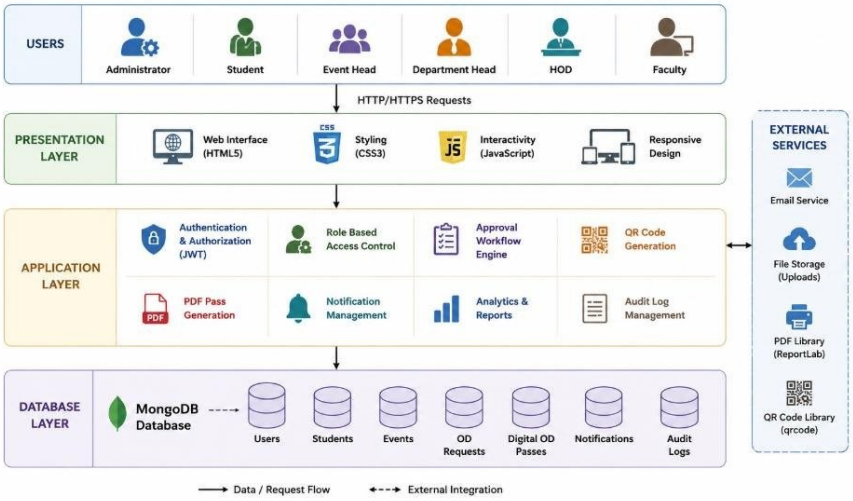


Figure 4.1 System Architecture of PassProof

PassProof follows a three-tier architecture of Presentation, Application, and Database layers, which ensures modularity, scalability, and easier maintenance.

Presentation Layer: A responsive interface built with HTML5, CSS3, and JavaScript that gives each role (Administrator, Student, Event Head, Department Head, HOD, Faculty) a dedicated dashboard.

Application Layer: Developed in Flask, it validates input, performs JWT authentication and authorization, executes business logic, and handles approval workflows, QR and PDF generation, notifications, conflict detection, and analytics.

Database Layer: MongoDB stores user information, OD requests, events, generated passes, notifications, and audit logs; its document-based design suits structured and semi-structured data.

5. MODULE DESCRIPTION

PassProof has six major modules that automate the OD approval workflow. Each is accessible only to authorized users through role-based authentication, which improves maintainability, security, and scalability.

5.1 Administrator Module

The central control unit of the system. The administrator creates user accounts individually or by bulk Excel upload (saving effort at the start of each academic year), resets passwords, activates or deactivates accounts, and monitors usage through dashboard statistics.

Key Features: Individual registration, bulk Excel upload, password reset, activate/deactivate accounts, role assignment, system statistics dashboard.

5.2 Student Module

Students apply for Individual OD by entering the reason, event date, duration, and supporting documents. The application goes to the Department Head, and students follow real-time status (pending, approved, or rejected) on their dashboard. After final HOD approval, a QR-enabled OD Pass is generated and can be downloaded as a PDF.

Key Features: Submit Individual OD, upload documents, view status, receive notifications, download QR OD Pass, view OD history.

5.3 Event Head Module

Designed for faculty or student coordinators organizing events. Instead of each participant applying separately, the Event Head creates an event and uploads the participant list, which is forwarded to the Department Head. Before submission, participants and event details can be edited or removed, simplifying large events involving multiple departments.

Key Features: Create events, add/edit participants, submit Event OD, track approval progress, view event history.

5.4 Department Head Module

Performs the first level of verification. The Department Head checks documents, validates student information, and evaluates each Individual and Event OD request, approving or rejecting with remarks. Approved requests move automatically to the HOD dashboard. Filters by department, event, or date simplify review.

Key Features: Review pending requests, verify documents, approve/reject with remarks, forward to HOD, view departmental history.

5.5 Head of the Department (HOD) Module

The final approval stage. The HOD reviews each request approved by the Department Head, verifies documents, and uploads the official approval letter. On approval, a unique QR-enabled digital OD Pass is generated for every student. The dashboard shows approved, rejected, and pending counts, and every action is recorded in the audit log.

Key Features: Final approval/rejection, upload approval documents, generate OD Pass, analytics dashboard, audit history.

5.6 Faculty Module

A read-only interface for verifying approved OD requests by register number search or QR scan. The system instantly shows the student's details, approval status, event information, date, and timing, removing dependence on physical letters and preventing unauthorized claims. Faculty cannot modify or approve records.

Key Features: Register number search, QR scan, verify digital pass, view approval details, instant verification.

5.7 Module Interaction

Students and Event Heads initiate OD requests, which the Department Head reviews and the HOD finally approves. The system then generates a digital OD Pass that faculty can verify, while the Administrator maintains user accounts and system operations throughout.

6. DATABASE DESIGN

PassProof uses MongoDB, a NoSQL document-oriented database, for its flexibility, scalability, and efficient handling of semi-structured data. Data is stored as JSON-like BSON documents without predefined table schemas, and relationships between collections are maintained using ObjectId references.

6.1 Database Overview

The database, named passproof_db, is the central repository for all application data. Every operation, including login, OD submission, approval, QR pass generation, and notification delivery, interacts with it, giving complete traceability and transparency.

6.2 Database Collections

The system has seven major collections:

Collection	Purpose
Users	Login credentials and profile data (name, email, role, password hash, department, status); primary authentication repository.
Students	Academic details (Register Number, department, year, section); linked to the Users collection.
Events	Details of symposiums, workshops, seminars, industrial visits, and placement activities created by Event Heads.
OD Records	Core collection storing every Individual and Event OD request with applicant details, type, reason, status, timestamps, and documents.
OD Pass	Digitally generated passes with Pass ID, QR Code, student information, approval details, and validity; used for faculty verification.
Notifications	System messages such as submission confirmation, approval, rejection, and pass generation alerts.
Audit Logs	Records logins, approvals, rejections, account changes, QR verifications, and admin actions with timestamps and user information.

6.3 Database Relationship

Logical relationships exist through ObjectId references: Users link to Students for authentication, Events link to OD Records for Event OD requests, approved OD Records generate OD Pass entries, and user activities produce Notifications and Audit Log entries. This allows efficient retrieval without complex joins.

6.4 Advantages of Using MongoDB

Flexible document-based storage, high scalability, faster read/write operations, efficient handling of dynamic data, easy integration with Flask through PyMongo, simple backup and restoration, better real-time performance, and reduced complexity compared to relational systems.

7. IMPLEMENTATION

The implementation converts the design into a working web application using Python Flask, MongoDB, HTML5, CSS3, and JavaScript. Authentication uses JWT, and QR Code technology enables instant verification of approved passes. The following screenshots show the major interfaces.

7.1 Login Page

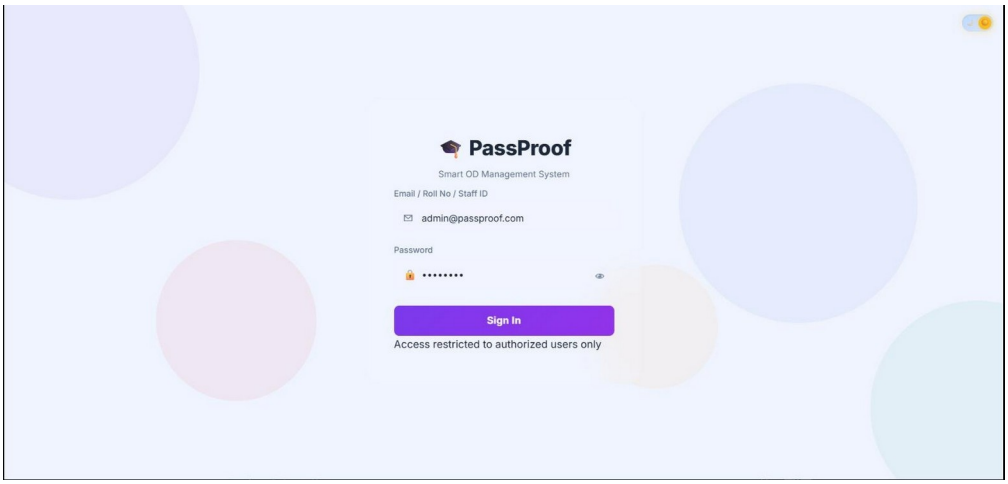


Figure 7.1 Login Page

The entry point for every user. Credentials are validated against the MongoDB database, a JWT is generated on success, and the user is redirected to the dashboard for their role. Invalid attempts are rejected with error messages.

7.2 Administrator Dashboard

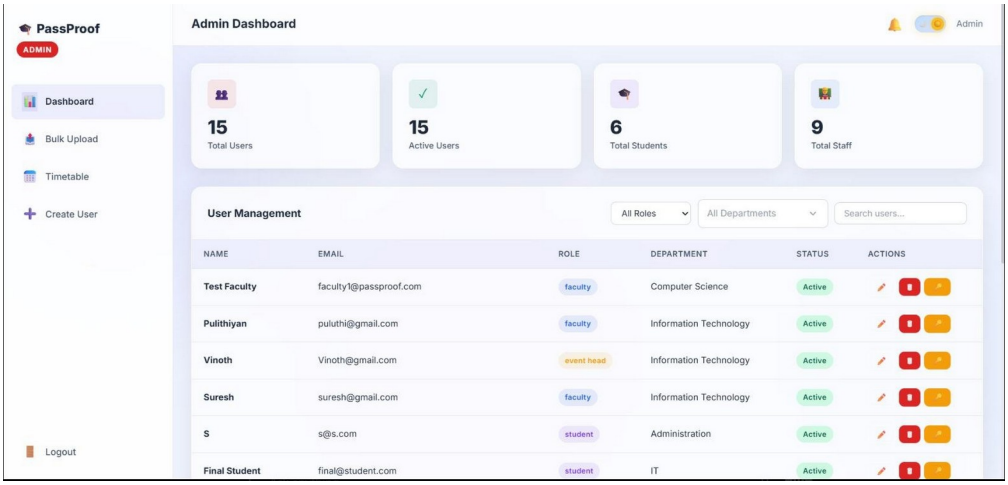


Figure 7.2 Administrator Dashboard

Shows statistics on registered users and overall activity. The administrator manages accounts, bulk uploads Excel files, resets passwords, and activates or deactivates users from this central interface.

7.3 Student Dashboard

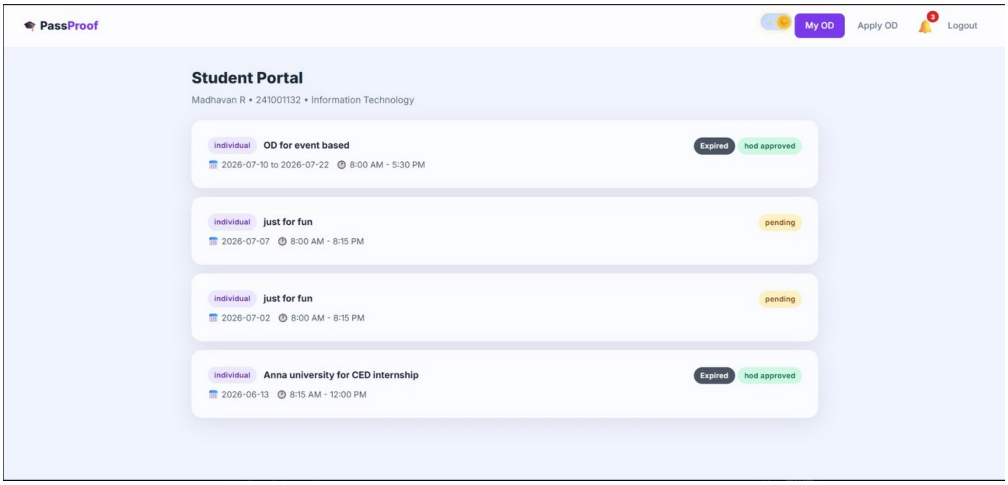


Figure 7.3 Student Dashboard

Gives students an overview of submitted OD requests and their current status, with approval history, notifications, and downloadable digital OD passes.

7.4 Individual OD Application

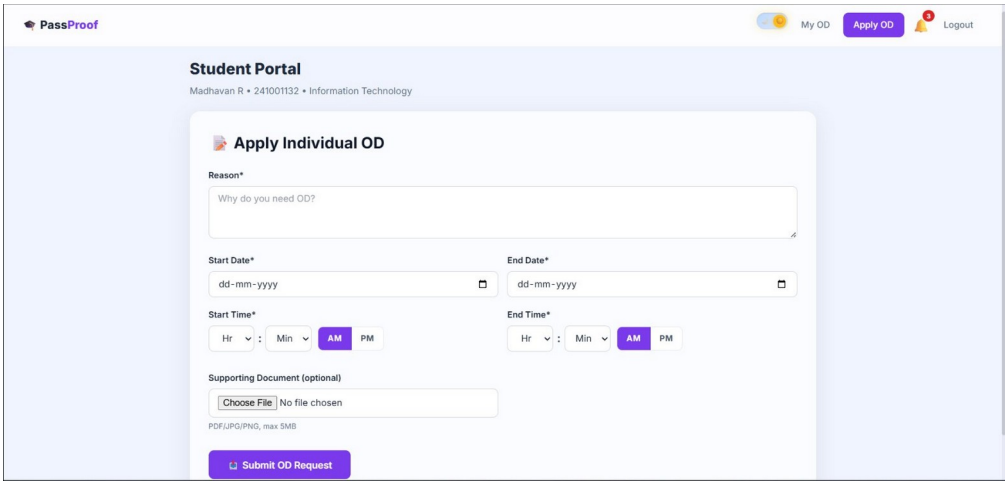


Figure 7.4 Individual OD Application

Students submit OD requests for internships, workshops, placement drives, industrial visits, and medical appointments by entering the reason, dates, times, and an optional supporting document. Mandatory fields are validated, and the request is forwarded to the Department Head.

7.5 HOD Approval Dashboard

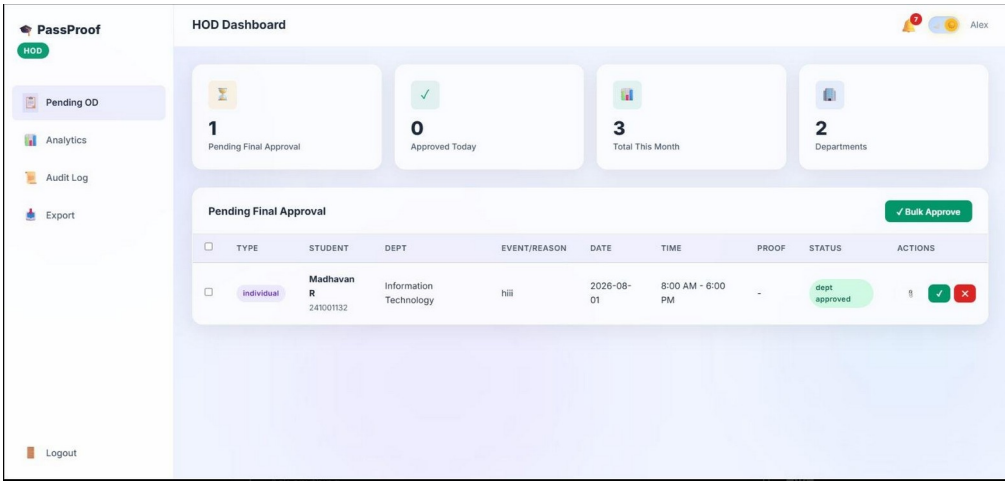


Figure 7.5 HOD Approval Dashboard

Lists requests approved by the Department Head and awaiting final approval. The HOD approves or rejects requests and uploads approval documents when needed; every action is recorded in the audit log.

7.6 Digital OD Pass

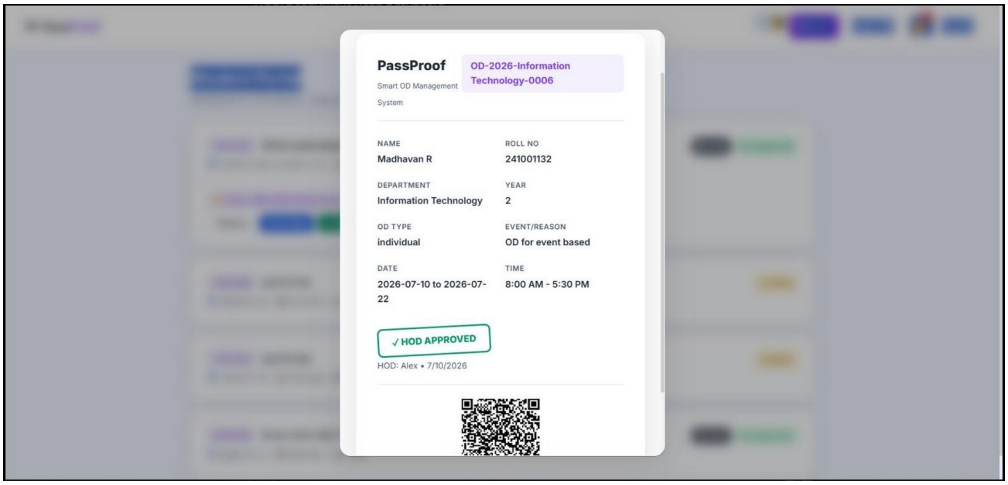


Figure 7.6 Generated Digital OD Pass with QR Code

After approval, the system generates a pass containing student information, event details, approval information, validity period, and a unique QR Code that faculty can scan to verify authenticity. Students can download it as a PDF.

7.7 Analytics Dashboard

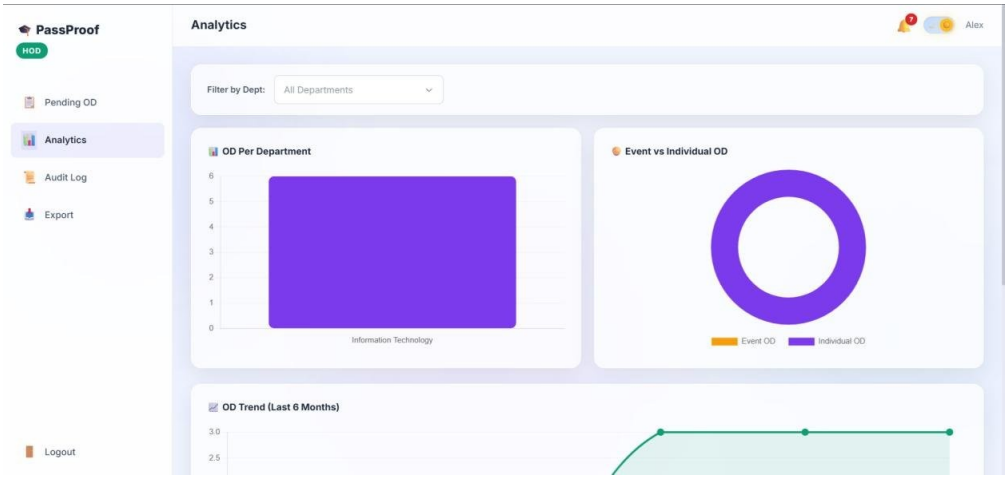


Figure 7.7 Analytics Dashboard

Provides graphical and numerical summaries of approved, pending, and rejected requests and registered users, helping authorities monitor the approval workflow through real-time reporting.

7.8 Summary

The implementation shows the successful integration of modern web technologies, secure authentication, and automated workflow management, effectively replacing manual OD procedures with a reliable digital platform.

8. SECURITY FEATURES

PassProof handles sensitive student information, approval records, and institutional data, so several mechanisms protect confidentiality, integrity, and authenticity.

8.1 JWT-Based Authentication: After login the server issues a unique JWT that authenticates every later request, preventing unauthorized access to protected resources.

8.2 Role-Based Access Control: Each user (Administrator, Student, Event Head, Department Head, HOD, Faculty) can access only the functions required for their role, so approvals and user management stay with authorized personnel.

8.3 Password Encryption: Passwords are hashed with bcrypt before storage and never kept in plain text; login compares the entered password with the hash, protecting credentials even if the database is compromised.

8.4 QR Code Verification: Each approved request carries a unique QR Code that faculty scan to verify it instantly, minimizing fake or modified passes.

8.5 Audit Logging: Logins, submissions, approvals, rejections, password resets, and QR verifications are logged, giving administrators complete traceability.

8.6 Session Management: Sessions are managed by JWT, and expired tokens, invalid sessions, and unauthorized attempts are automatically rejected.

9. TESTING

Functional, integration, and user acceptance testing were carried out to verify that every module performs correctly under different operating conditions.

9.1 Functional Testing

Test Case	Expected Result	Status
User Login	Successful authentication with valid credentials	Pass
Student Registration	Student account created successfully	Pass
Individual OD Submission	Request stored successfully	Pass
Event OD Submission	Event request created successfully	Pass
Department Head Approval	Request forwarded to HOD	Pass
HOD Approval	Digital OD Pass generated	Pass
QR Verification	Valid student details displayed	Pass
PDF Generation	Digital pass downloaded successfully	Pass

9.2 Integration Testing

Login correctly authenticated users before granting dashboard access. Requests moved correctly from the Student Module to the Department Head Module and then the HOD Module, and approvals triggered QR Code generation, PDF generation, notifications, and database updates without errors.

9.3 User Acceptance Testing

Students, faculty, and administrators performed their assigned operations and confirmed the application was easy to use and met the requirements. Feedback indicated reduced paperwork, simpler approvals, and better transparency than manual methods.

9.4 Testing Summary

All major functionalities operate correctly and reliably, satisfying the functional requirements while maintaining security, performance, and usability.

10. CONCLUSION AND FUTURE ENHANCEMENTS

10.1 Conclusion

PassProof – Smart OD Management System has been designed and developed as a secure, reliable, and user-friendly web application that replaces paper-based OD approval with a centralized digital platform, reducing administrative workload and improving transparency across the approval lifecycle.

Dedicated dashboards ensure each role performs only permitted operations, while JWT-based role authentication and bcrypt password encryption protect sensitive information. A major achievement is QR Code-based OD verification, which lets faculty verify passes instantly without physical documents. Real-time tracking, audit logging, automated PDF generation, notifications, and analytics further simplify administration.

Built with Python Flask, MongoDB, HTML5, CSS3, and JavaScript, the system is scalable and maintainable, and it fulfills the objective of providing a secure, transparent, and efficient OD management solution for students, faculty, and administrators.

10.2 Future Enhancements

- Android and iOS mobile applications.
- Integration with the College ERP for automatic attendance synchronization.
- Email and SMS notifications, and real-time push notifications using Firebase Cloud Messaging.
- Digital signature verification during final approval.
- Cloud deployment for better availability and scalability.
- AI to predict approval trends and identify unusual request patterns.
- Face recognition or biometric verification for additional authentication.
- Multi-college support and advanced interactive analytics dashboards.

11. SUSTAINABLE DEVELOPMENT GOALS (SDG) MAPPING

11.1 SDG Mapping Table

IT23723 – Internship II Project Title: PASSPROOF – SMART OD MANAGEMENT SYSTEM

SDG Goal	Contribution of PassProof
SDG 4 – Quality Education	Supports students in academic and co-curricular activities through an efficient digital approval system.
SDG 9 – Industry, Innovation and Infrastructure	Implements a modern web platform with secure authentication, QR verification, and workflow automation.
SDG 12 – Responsible Consumption and Production	Eliminates paper-based OD applications by maintaining digital records and approvals.

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APPENDIX

The appendix contains supplementary material supporting the development of PassProof. The six project modules and the database collections are described in Chapters 5 and 6.

A.1 Project Configuration Files

- .env.example – environment variables for configuring the application without exposing sensitive credentials.
- requirements.txt – Python libraries and package versions required to run the application.
- config.py – database connection settings, JWT configuration, and secret keys.
- app.py – main entry point of the Flask application.
- README.md – installation instructions, project overview, and usage guidelines.